

DNS Server

BIND9 · Linux System & Network Administration

1. Service Fundamentals

DNS (Domain Name System) is a hierarchical, distributed database that translates human-readable names into IP addresses (forward lookup) and optionally IP addresses into names (reverse lookup). It underpins almost every network service.

Item	Details
Package	bind9 · bind9utils · dnsutils
Port	UDP/TCP 53
Main Config	/etc/bind/named.conf · named.conf.options · named.conf.local
Zone Files	/etc/bind/db.*
Service Unit	named.service / bind9.service
Record Types	A, AAAA, CNAME, MX, NS, PTR, SOA, TXT

2. Architecture

Client → Recursive resolver → Root → TLD → Authoritative nameserver. Authoritative servers hold the definitive zone data; recursive servers perform lookups and cache answers. Hierarchy: Root (.) → TLD (.com) → Domain (example) → Host (www).

3. Installation

Update package list and install BIND9:

Command:

```
sudo apt update
```

Command:

```
sudo apt install bind9 bind9utils dnsutils
```

Enable and start the service:

Command:

```
sudo systemctl enable --now named
```

4. Configuration

Example lab domain: company.local · Server IP: 192.168.100.10

Define the zones:

Command:

```
sudo nano /etc/bind/named.conf.local

zone "company.local" {
    type master;
    file "/etc/bind/db.company.local";
};

zone "100.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.100";
};
```

Create the forward zone file:

Command:

```
sudo nano /etc/bind/db.company.local

$TTL 86400
@ IN SOA ns1.company.local. admin.company.local. (
2026092101 ; Serial
3600 ; Refresh
1800 ; Retry
604800 ; Expire
86400 ) ; Minimum
@ IN NS ns1.company.local.
```

```
ns1 IN A 192.168.100.10
www IN A 192.168.100.20
mail IN A 192.168.100.30
```

Validate and reload:

Command:

```
sudo named-checkconf
```

Command:

```
sudo named-checkzone company.local /etc/bind/db.company.local
```

Command:

```
sudo systemctl reload named
```

5. Verification

Query the forward record:

Command:

```
dig @192.168.100.10 www.company.local +short
```

Query a reverse record:

Command:

```
dig @192.168.100.10 -x 192.168.100.20 +short
```

Confirm the service is listening:

Command:

```
sudo ss -ltnp | grep :53
```

6. Configuration Reference

File / Record	Purpose
named.conf.options	Global options (recursion, listen-on)
named.conf.local	Zone definitions
SOA	Start of Authority — zone metadata & serial
NS	Nameserver for the zone
A / AAAA	IPv4 / IPv6 address
PTR	Reverse (IP → name)
MX	Mail exchanger
CNAME	Alias to another name

7. Troubleshooting

Problem	Check	Solution
named fails to start	named-checkconf	Fix syntax in named.conf*
Zone not loading	named-checkzone · logs	Correct SOA serial, NS, file path
NXDOMAIN answer	dig · zone file	Add missing A/PTR records; reload
Clients use wrong DNS	resolv.conf · DHCP options	Point clients to the internal DNS IP

8. Security

- Disable recursion on authoritative-only servers (recursion no;).
- Restrict allow-query and allow-transfer to trusted networks.
- Use TSIG or ACLs for zone transfers.
- Keep BIND updated; never run an open recursive resolver on the public Internet.

9. Real-World Scenario

A company runs an internal domain company.local. BIND is the master for the zone. Workstations receive the DNS server address via DHCP. Web, mail and application servers are registered with A records; a reverse zone is maintained for logging.

10. Essential Commands

Command	Purpose
sudo apt install bind9 bind9utils dnstools	Install BIND9

Command	Purpose
sudo named-checkconf	Validate configuration
sudo named-checkzone ZONE FILE	Validate a zone file
sudo systemctl reload named	Apply zone changes
dig @server name	Query a specific server
dig -x IP	Reverse lookup

11. Important Points (Interview / Viva)

1. DNS maps names ↔ IP addresses · Port UDP/TCP 53.
2. Package: bind9 · Config under /etc/bind/.
3. SOA serial must increase on every zone change.
4. Validate with named-checkconf and named-checkzone before reload.
5. A = forward, PTR = reverse, MX = mail, NS = nameserver.